01 - EDA Exploratory Data Analysis - data splits

In this notebook, we are going to:

- extract text from transcribed txt files
- map the text to its corresponding labels (AD/CN)
- do EDA (Exploratory Data Analysis) on the train set

We will then split the train set into 2 sets: train and development, with the ratio 50/50

In [1]:
```
# import required packages
import pandas as pd
import os
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

In [2]:
```
# labels
# AD = Alzheimer's Disease (positive class)
# CN = Cognitively Normal (negative class)
```

In [3]:
```
# please point to correct downloaded transcribed folders of the original dataset
data_path = "../../../data/"
path_ad = data_path + "Transcribed Data-2024-Feb-13/Diagnosis Training Data/ad tran
path_cn = data_path + "Transcribed Data-2024-Feb-13//Diagnosis Training Data/cn tra
```

In [4]:
```
train_data = {"ad":[],
              "cn":[]
             }
```

In [5]:
```
def read_data(path):
    """
        get data from "txt" files from a specified folder (ad/cn)
        returns list of text (strings)
    """
    result = []
    # get only txt files
    files = [item
             for item in os.listdir(path)
             if item.endswith("txt")
            ]

    for fname in files:
        print("Processing files =", fname) #, end="\r")
        with open(path + fname, encoding="utf-8") as f:
            lines = f.readlines()
            print("Line len =", len(lines)) #, end="\r")
            text = "\n".join(lines)
            result.append(text)
```

```
    return result

train_data["ad"] = read_data(path_ad)
train_data["cn"] = read_data(path_cn)
```

```
Processing files = adrso024.txt
Line len = 23
Processing files = adrso025.txt
Line len = 13
Processing files = adrso027.txt
Line len = 9
Processing files = adrso028.txt
Line len = 17
Processing files = adrso031.txt
Line len = 29
Processing files = adrso032.txt
Line len = 5
Processing files = adrso033.txt
Line len = 7
Processing files = adrso035.txt
Line len = 9
Processing files = adrso036.txt
Line len = 21
Processing files = adrso039.txt
Line len = 9
Processing files = adrso043.txt
Line len = 27
Processing files = adrso045.txt
Line len = 35
Processing files = adrso046.txt
Line len = 13
Processing files = adrso047.txt
Line len = 3
Processing files = adrso049.txt
Line len = 15
Processing files = adrso053.txt
Line len = 21
Processing files = adrso054.txt
Line len = 21
Processing files = adrso055.txt
Line len = 9
Processing files = adrso056.txt
Line len = 29
Processing files = adrso059.txt
Line len = 3
Processing files = adrso060.txt
Line len = 19
Processing files = adrso063.txt
Line len = 9
Processing files = adrso068.txt
Line len = 9
Processing files = adrso070.txt
Line len = 3
Processing files = adrso071.txt
Line len = 3
Processing files = adrso072.txt
Line len = 1
Processing files = adrso074.txt
Line len = 3
Processing files = adrso075.txt
Line len = 3
```

```
Processing files = adrso077.txt
Line len = 1
Processing files = adrso078.txt
Line len = 11
Processing files = adrso089.txt
Line len = 1
Processing files = adrso090.txt
Line len = 11
Processing files = adrso092.txt
Line len = 1
Processing files = adrso093.txt
Line len = 1
Processing files = adrso098.txt
Line len = 5
Processing files = adrso106.txt
Line len = 3
Processing files = adrso109.txt
Line len = 3
Processing files = adrso110.txt
Line len = 5
Processing files = adrso112.txt
Line len = 5
Processing files = adrso116.txt
Line len = 9
Processing files = adrso122.txt
Line len = 7
Processing files = adrso123.txt
Line len = 5
Processing files = adrso125.txt
Line len = 1
Processing files = adrso126.txt
Line len = 5
Processing files = adrso128.txt
Line len = 5
Processing files = adrso130.txt
Line len = 1
Processing files = adrso134.txt
Line len = 5
Processing files = adrso138.txt
Line len = 5
Processing files = adrso141.txt
Line len = 1
Processing files = adrso142.txt
Line len = 3
Processing files = adrso144.txt
Line len = 41
Processing files = adrso187.txt
Line len = 1
Processing files = adrso188.txt
Line len = 19
Processing files = adrso189.txt
Line len = 5
Processing files = adrso190.txt
Line len = 3
Processing files = adrso192.txt
Line len = 15
```

```
Processing files = adrso197.txt
Line len = 5
Processing files = adrso198.txt
Line len = 39
Processing files = adrso200.txt
Line len = 13
Processing files = adrso202.txt
Line len = 7
Processing files = adrso205.txt
Line len = 3
Processing files = adrso206.txt
Line len = 5
Processing files = adrso209.txt
Line len = 17
Processing files = adrso211.txt
Line len = 3
Processing files = adrso212.txt
Line len = 13
Processing files = adrso215.txt
Line len = 5
Processing files = adrso216.txt
Line len = 17
Processing files = adrso218.txt
Line len = 7
Processing files = adrso220.txt
Line len = 29
Processing files = adrso222.txt
Line len = 1
Processing files = adrso223.txt
Line len = 7
Processing files = adrso224.txt
Line len = 1
Processing files = adrso228.txt
Line len = 3
Processing files = adrso229.txt
Line len = 9
Processing files = adrso232.txt
Line len = 3
Processing files = adrso233.txt
Line len = 7
Processing files = adrso234.txt
Line len = 15
Processing files = adrso236.txt
Line len = 5
Processing files = adrso237.txt
Line len = 3
Processing files = adrso244.txt
Line len = 1
Processing files = adrso245.txt
Line len = 13
Processing files = adrso246.txt
Line len = 3
Processing files = adrso247.txt
Line len = 7
Processing files = adrso248.txt
Line len = 3
```

```
Processing files = adrso249.txt
Line len = 17
Processing files = adrso250.txt
Line len = 1
Processing files = adrso253.txt
Line len = 5
Processing files = adrso002.txt
Line len = 1
Processing files = adrso003.txt
Line len = 1
Processing files = adrso005.txt
Line len = 1
Processing files = adrso007.txt
Line len = 1
Processing files = adrso008.txt
Line len = 1
Processing files = adrso010.txt
Line len = 1
Processing files = adrso012.txt
Line len = 1
Processing files = adrso014.txt
Line len = 1
Processing files = adrso015.txt
Line len = 5
Processing files = adrso016.txt
Line len = 1
Processing files = adrso017.txt
Line len = 7
Processing files = adrso018.txt
Line len = 1
Processing files = adrso019.txt
Line len = 3
Processing files = adrso021.txt
Line len = 5
Processing files = adrso022.txt
Line len = 1
Processing files = adrso023.txt
Line len = 11
Processing files = adrso148.txt
Line len = 7
Processing files = adrso151.txt
Line len = 5
Processing files = adrso152.txt
Line len = 3
Processing files = adrso153.txt
Line len = 7
Processing files = adrso154.txt
Line len = 5
Processing files = adrso156.txt
Line len = 5
Processing files = adrso157.txt
Line len = 7
Processing files = adrso158.txt
Line len = 3
Processing files = adrso159.txt
Line len = 3
```

```
Processing files = adrso160.txt
Line len = 1
Processing files = adrso161.txt
Line len = 1
Processing files = adrso162.txt
Line len = 3
Processing files = adrso164.txt
Line len = 5
Processing files = adrso165.txt
Line len = 5
Processing files = adrso167.txt
Line len = 5
Processing files = adrso168.txt
Line len = 5
Processing files = adrso169.txt
Line len = 3
Processing files = adrso170.txt
Line len = 1
Processing files = adrso172.txt
Line len = 7
Processing files = adrso173.txt
Line len = 1
Processing files = adrso177.txt
Line len = 3
Processing files = adrso178.txt
Line len = 3
Processing files = adrso180.txt
Line len = 3
Processing files = adrso182.txt
Line len = 5
Processing files = adrso183.txt
Line len = 5
Processing files = adrso186.txt
Line len = 3
Processing files = adrso257.txt
Line len = 5
Processing files = adrso259.txt
Line len = 3
Processing files = adrso260.txt
Line len = 3
Processing files = adrso261.txt
Line len = 3
Processing files = adrso262.txt
Line len = 3
Processing files = adrso263.txt
Line len = 13
Processing files = adrso264.txt
Line len = 9
Processing files = adrso265.txt
Line len = 17
Processing files = adrso266.txt
Line len = 3
Processing files = adrso267.txt
Line len = 25
Processing files = adrso268.txt
Line len = 5
```

```
Processing files = adrso270.txt
Line len = 5
Processing files = adrso273.txt
Line len = 3
Processing files = adrso274.txt
Line len = 3
Processing files = adrso276.txt
Line len = 3
Processing files = adrso277.txt
Line len = 3
Processing files = adrso278.txt
Line len = 1
Processing files = adrso280.txt
Line len = 17
Processing files = adrso281.txt
Line len = 3
Processing files = adrso283.txt
Line len = 3
Processing files = adrso285.txt
Line len = 3
Processing files = adrso286.txt
Line len = 1
Processing files = adrso289.txt
Line len = 5
Processing files = adrso291.txt
Line len = 1
Processing files = adrso292.txt
Line len = 3
Processing files = adrso296.txt
Line len = 11
Processing files = adrso298.txt
Line len = 7
Processing files = adrso299.txt
Line len = 5
Processing files = adrso300.txt
Line len = 3
Processing files = adrso302.txt
Line len = 3
Processing files = adrso307.txt
Line len = 1
Processing files = adrso308.txt
Line len = 5
Processing files = adrso309.txt
Line len = 1
Processing files = adrso310.txt
Line len = 3
Processing files = adrso312.txt
Line len = 5
Processing files = adrso315.txt
Line len = 5
Processing files = adrso316.txt
Line len = 5
```

In [6]:

```python
# double check data
assert len(train_data["ad"]) == 87
assert len(train_data["cn"]) == 79
```

```
assert train_data["ad"][0].startswith("The cookie jar and the little")
assert train_data["ad"][1].startswith("Tell me everything that's going on")
assert train_data["cn"][0].startswith("Okay, I see a mom doing")
assert train_data["cn"][1].startswith("Starting from the left I see")
print("Passed")
```

```
Passed
```

In [7]:
```
# take a look at the data
print(train_data["ad"][0][:100])
```

```
The cookie jar and the little
```

```
boy about down on the floor
```

```
girls
```

```
Girls anyway the housewife
```

In [8]:
```
# take a look at the data
print(train_data["cn"][0][:100])
```

```
Okay, I see a mom doing the dishes and the waters overflowing from the sink and she
seems to be comp
```

In [9]:
```
# build word-clouds
def make_wordcloud(text_list):
    """
    display a wordcloud based on the text list
    parameters:
        text_list: list of text (paragraphs)
    returns:
        list of frequent words
    """
    text = "\n".join(text_list).lower()
    wordcloud = WordCloud().generate(text)

    # process text and return counts of frequent words
    result = sorted(WordCloud().process_text(text).items(),
                    key=lambda item: item[1],
                    reverse=True
                   )


    plt.imshow(wordcloud)
    plt.axis("off")
    plt.show()

    return result
```

In [10]:
```
# see the most frequent words of AD (positive) group
ad_top_words = make_wordcloud(train_data["ad"])
print(ad_top_words[:50])
ad_top_words_set = set([w
                        for w, count in ad_top_words[:200]
                       ])
```




```
[('picture', 102), ('going', 100), ('water', 98), ('okay', 80), ('tell', 69), ('cook
ie', 63), ('cookie jar', 59), ('see', 58), ('boy', 57), ('anything', 56), ('everythi
ng', 52), ('dishes', 52), ('mother', 46), ('sink', 45), ('look', 44), ('see going',
42), ('girl', 41), ('oh', 41), ('one', 38), ('floor', 36), ('well', 36), ('running',
33), ('happening', 32), ('kid', 30), ('hand', 28), ('right', 27), ('know', 26), ('ge
tting', 25), ('yeah', 24), ('want', 24), ('stool', 24), ('thing', 23), ('see happeni
ng', 23), ('trying', 22), ('fall', 22), ('something', 21), ('window', 21), ('think',
21), ('washing dishes', 21), ('falling', 20), ('say', 18), ('now', 18), ('plate', 1
7), ('action', 17), ('little girl', 17), ('okay good', 17), ('still', 16), ('go', 1
5), ('washing', 15), ('chair', 15)]
```

In [11]:
```
# see the most frequent words of CN (negative) group
cn_top_words = make_wordcloud(train_data["cn"])
print(cn_top_words[:50])
cn_top_words_set = set([w
                        for w, count in cn_top_words[:200]
                       ])
```




```
[('water', 102), ('cookie', 97), ('sink', 85), ('mother', 75), ('cookie jar', 75),
('boy', 71), ('window', 69), ('stool', 54), ('see', 53), ('picture', 53), ('going',
53), ('one', 52), ('look', 52), ('dishes', 51), ('girl', 46), ('running', 46), ('eve
rything', 45), ('okay', 41), ('reaching', 40), ('getting', 38), ('tell', 38), ('floo
r', 37), ('hand', 35), ('drying', 34), ('overflowing', 33), ('right', 32), ('curtai
n', 31), ('standing', 30), ('open', 29), ('falling', 28), ('fall', 26), ('sister', 2
6), ('want', 24), ('little girl', 24), ('action', 23), ('plate', 22), ('anything', 2
1), ('see going', 21), ('know', 20), ('looking', 20), ('kitchen', 20), ('trying', 2
0), ('outside', 19), ('dish', 19), ('another', 19), ('store', 19), ('washing dishe
s', 18), ('onto', 17), ('thing', 17), ('little boy', 17)]
```

Observations

We can see that there are some words that are common in both categories (AD/CN):

- cookie, cookie jar, water, sink, boy, girl, etc

In [12]:
```
# observe the difference in 2 word set
ad_unique = ad_top_words_set - cn_top_words_set
cn_unique = cn_top_words_set - ad_top_words_set
print("AD unique words =\n", ad_unique)
print("----------------")
print("CN unique words =\n", cn_unique)
```

```
AD unique words =
 {'touching', 'made', 'rather', 'whatever', 'put', 'fell', 'handles', 'pipe', 'told', 'find', 'whether', 'cooking', 'always', 'two cup', 'suppose', 'shoes', 'ha', 'lot', 'okay good', 'yes', 'us', 'sorry', 'upset', 'eat', 'lady', 'stealing', 'watching', 'neck', 'small', 'different', 'love', 'ladder', 'step', 'read', 'saucer', 'try', 'next', 'guy', 'stove', 'stand', 'pretty', 'goes', 'telling', 'whoa whoa', 'young', 'hard', 'spill', 'slipping', 'working', 'almost', 'home', 'somebody', 'assume', 'spilled', 'share', 'problem', 'cup', 'stone', 'break', 'third', 'bushes', 'wash', 'curtains', 'new', 'breeze', 'broke', 'call', 'given', 'gotta', 'taken', 'underneath', 'grabbing'}
---------------
CN unique words =
 {'either', 'face', 'doors open', 'mouth', 'crash', 'room', 'wind', 'towel', 'show', 'course', 'attention', 'behind', 'place', 'son', 'letting', 'accident', 'ground', 'sitting', 'shelf', 'walking', 'scene', 'tipping', 'though', 'oblivious', 'even', 'finger', 'grab', 'lid', 'drawing', 'wet', 'wrong', 'blowing', 'paying', 'tilting', 'nothing', 'flowing', 'two cups', 'foot', 'path', 'dress', 'come', 'paying attention', 'giving', 'dried', 'appear', 'two children', 'cupboard door', 'summer', 'start', 'tree', 'tie', 'part', 'stealing cookies', 'walkway', 'putting', 'corner', 'curtain', 'splashing', 'mama', 'shoe', 'faucet', 'left hand', 'garden', 'stepping', 'dry', 'turning', 'brother', 'full', 'lip', 'obviously', 'wiping', 'handing'}
```

Observation

Some words that frequently appear in AD (positive) group, but not so frequently in CN (negative) groups:

- almost, okay good, suppose, somebody, broke, etc

These words tend to be generic (not with very specific meaning)

Some words that frequently appear in CN (negative) group, but not so frequently in AD (positive) groups:

- stepping, dress, splashing, grab, accident, etc

These words tend to be more specific and detailed.

In [13]:
```python
# combine data into a dataframe, for export to a csv for further processing
train_dict = {
    "text": [],  # the transcript
    "target": [] # the target: positive (AD), negative (CN)
}
for item in train_data["ad"]:
    train_dict["text"].append(item)
    train_dict["target"].append("AD")

for item in train_data["cn"]:
    train_dict["text"].append(item)
    train_dict["target"].append("CN")

print("Text count =", len(train_dict["text"]))
print("Target count =", len(train_dict["target"]))
```

```
Text count = 166
Target count = 166
```

In [14]:
```
train_df = pd.DataFrame(train_dict)
print(len(train_df))
train_df.head()
```

```
166
```

Out[14]:

	text	target
0	The cookie jar and the little\n\n\n\nboy about...	AD
1	Tell me everything that's going on\n\n\n\nthe ...	AD
2	As a picture little girl and a little boy stan...	AD
3	She will find her mother worship trying to dis...	AD
4	What do you see going on\n\n\n\nthe boys on a ...	AD

In [15]:
```
train_df.describe()
```

Out[15]:

	text	target
count	166	166
unique	166	2
top	The cookie jar and the little\n\n\n\nboy about...	AD
freq	1	87

We can see we have total of 166 examples, with 87 positive (AD) and 79 negative (CN). The train_set is relatively balanced.

Now we can split the data into train and dev set

In [16]:
```
X_train, X_dev, y_train, y_dev  = train_test_split(train_df["text"],
                                                   train_df["target"],
                                                   test_size=0.5, # split 50/50 for
                                                   random_state=42
                                                  )
print("X_train len =", len(X_train))
print("y_train len =", len(y_train))
print("X_dev len =", len(X_dev))
print("y_dev len =", len(y_dev))
```

```
X_train len = 83
y_train len = 83
X_dev len = 83
y_dev len = 83
```

In [17]:
```
# write to csv
path = data_path + "splits/"
train_set_df = pd.DataFrame({"text":X_train,
```

```
                                "target":y_train
                               })

dev_set_df = pd.DataFrame({"text":X_dev,
                           "target":y_dev
                          })
```

In [18]:
```
train_set_df.head()
```

Out[18]:

	text	target
79	I think you see going on in that picture. We'l...	AD
39	I'm gonna give you a picture that has a lot of...	AD
154	Ah the sinks running over waters come on all f...	CN
86	Well, my boys trying to get some cookies and t...	AD
117	I'd like you to do is just look at the picture...	CN

In [19]:
```
dev_set_df.head()
```

Out[19]:

	text	target
100	Everything you see going on in this picture, g...	CN
136	Now you just look at the picture. Okay, this i...	CN
78	Tell me everything that you see going on in th...	AD
56	Girls and boys getting her one that stores sto...	AD
97	All right, we have a few more tasks to go thro...	CN

In [20]:
```
print("Writing to csv files...")

train_set_df.to_csv(path + "train_50pc.csv",
                    index=False)
dev_set_df.to_csv(path + "dev_50pc.csv",
                  index=False)
print("Completed")
```

```
Writing to csv files...
Completed
```

In [ ]: